DDPO Reward Components — Mathematical Specification

`ddpm-jax` · physics-aware finetuning of the SR diffusion model
Implementation: `src/rewards.py` · Calibration: `base_results/reward_calibration.ipynb` · 6 July 2026

Purpose. Exact definitions of the four reward families as implemented, with every deliberate modelling choice flagged for theoretical review. Line references point at the implementation so math-vs-code can be checked directly. Calibration evidence (monotonicity ladders, hackability probes, cross-Re tables) lives in the notebook; this document is the theory.

1. Setup and conventions

Every reward maps a **normalized vorticity triplet** to a per-sample scalar distance. With ω the physical vorticity on an $N \times N$ grid ($N = 256$) over the torus $(0, 2\pi)^2$:

$$x = \frac{\omega - \mu}{\sigma} \in \mathbb{R}^{N \times N \times 3}, \quad \mu = 0, \quad \sigma = 4.7988$$

σ, μ are the model's training statistics (Re$=1000$, first-32-sequence split) and are used for **all** regimes — the model only ever sees this normalization; the regime anchors (§2) absorb the residual mismatch. Channels are consecutive frames $(\omega_{t-1}, \omega_t, \omega_{t+1})$. Statistical rewards use only the middle frame x_{mid}; the PDE residual uses all three. Rewards are evaluated on the final denoised sample x_0 only — DDPO needs scalars, not reward gradients.

Spectral conventions (identical to `lambda_sweep_re1000.spectrum_fn` and `docs/energy_spectrum_metric.md`). Unnormalized 2-D DFT $\hat{x}(k) = \sum_x x(x) e^{-ik \cdot x}$ with integer wavenumbers $k = (k_x, k_y)$, and radial shell index $\kappa(k) = \text{round}(|k|)$. The **enstrophy (vorticity-power) spectrum** of the normalized middle frame is the shell sum

$$E(k) = \sum_{\kappa(k') = k} |\hat{x}_{\text{mid}}(k')|^2, \quad k = 0, ..., N/2 - 1 = 127$$

Parseval in the numpy convention, $\sum_k |\hat{x}(k)|^2 = N^2 \sum_x x(x)^2$, gives $\sum_k E(k) = N^4 \cdot \langle x_{\text{mid}}^2 \rangle$ (with $\langle \cdot \rangle$ the spatial mean), so $E(k)$ is the total normalized enstrophy decomposed by scale. The kinetic-energy spectrum is the same data reweighted by $1/(2k^2)$; the enstrophy convention is used because it weights exactly the high-k band where over-smoothing lives.

Each component below is a distance $d(x) \geq 0$, lower is better; the combined DDPO reward negates them (§7).

2. Regime anchors (`compute_regime_stats`, rewards.py:61)

All rewards are **reference-free with respect to paired ground truth**: they are anchored to regime-level statistics computed once from any M frames $\{x_m\}$ of the target regime (train split, or a generated run at the target Re). This is what allows the same rewards to drive finetuning toward regimes with no paired HR data, and prevents the reward from collapsing back into the MSE objective whose high-k blindness is the thing being fixed.

$$\mathrm{spec}_{\mathrm{ref}}(k) = \frac{1}{M}\sum_{m=1}^{M} E_m(k) \qquad \text{(arithmetic mean — stored, not used)}$$

$$\Lambda(k) := \log \mathrm{spec}_{\mathrm{ref}}(k) = \frac{1}{M}\sum_{m=1}^{M} \ln E_m(k) \qquad \text{(geometric-mean anchor — used)}$$

$$\mathcal{E}_{\mathrm{ref}} = \frac{1}{M}\sum_{m=1}^{M} \langle x_m^2 \rangle \qquad \text{(total-enstrophy anchor)}$$

$$q_{\mathrm{ref}}(j) = \frac{1}{M}\sum_{m=1}^{M} F_m^{-1}\left(\frac{j}{Q-1}\right), \quad j = 0, \ldots, Q-1, \quad Q = 257$$

where F_m^{-1} is frame m's empirical quantile function (linear interpolation).

Why the geometric mean for the spectral anchor (first-pass calibration finding): per-shell frame spectra are approximately log-normally distributed, so the arithmetic mean lies **above** the typical frame's spectrum. With an arithmetic anchor, a slightly blurred frame scored **better** than GT (blur $\sigma_b = 0.5$ gave $d_{\mathrm{spec}} = 0.09$ against a GT floor of 0.24) — the reward paid the model to smooth. The geometric mean is the minimizer of the expected squared log-distance, so the typical frame sits at the floor. Challenge points: a per-shell median anchor, or per-shell variance weighting (Mahalanobis in log space).

3. Spectrum matching — d_{spec}, $d_{\mathrm{spec,hi}k}$ (rewards.py:106)

$$d_{\mathrm{spec}}(x) = \frac{1}{|B|}\sum_{k\in B}\left[\ln \tilde{E}(k) - \Lambda(k)\right]^2, \qquad \tilde{E}(k) = \max\left(E(k),\ \varepsilon e^{\Lambda(k)}\right), \quad \varepsilon = 10^{-6}$$

$$B = \{1, \ldots, 95\} \quad \text{for} \quad d_{\mathrm{spec}}, \qquad B = \{32, \ldots, 95\} \quad \text{for} \quad d_{\mathrm{spec,hi}k}$$

- **Log space:** $E(k)$ falls ~ 3 decades across the band; a linear metric would be dominated by the first few shells and blind at high k — reproducing exactly the failure mode of L2. In log space every decade of the enstrophy cascade counts equally: this is a per-scale **relative** energy error.
- **Band:** $k = 0$ excluded (mean mode, ≈ 0 on normalized data); $k \geq 96$ excluded because GT energy $\to 0$ there and log-ratios become noise. The high-k band starts at $k = 32$, well above the forcing scale $k_f = 4$ — the "deficit band" of the OOD analysis.
- **Relative floor ε:** a hard-zeroed shell contributes a bounded $(\ln 10^{-6})^2 \approx 191$ instead of an epsilon-driven blow-up.
- **Blind spot (by construction, proven by the scramble probe):** phases. Any field with the reference spectrum — including spectrum-matched noise — sits at the GT floor. Matching marginal spectra is necessary, not sufficient.

Relation to the tracked OOD metric: hi-k retention $R_{k\geq 32} = \left(\sum_{k\geq 32} E_{\mathrm{recon}}(k)\right) / \left(\sum_{k\geq 32} E_{\mathrm{GT}}(k)\right)$ is a GT-paired linear band ratio; $d_{\mathrm{spec,hi}k}$ is its reference-free log-space counterpart. Calibration: rank correlation within-input ≈ -0.7, model-level -1.0.

4. Energy conservation — d_{energy} (rewards.py:128)

$$d_{\mathrm{energy}}(x) = \left[\ln\langle x_{\mathrm{mid}}^2\rangle - \ln \mathcal{E}_{\mathrm{ref}}\right]^2$$

Total normalized enstrophy against the regime mean, as a squared log-ratio (symmetric: over- and under-energized cost the same; a factor $e^{\pm 1}$ costs 1). By Parseval this is the k-integral of the spectrum, so it is scale-blind — it pins the overall level while d_{spec} fixes the distribution across scales.

Note this is **enstrophy** conservation, not kinetic energy: the KE analogue would anchor $\sum_k E(k)/(2k^2)$, which is low-k weighted. Enstrophy was chosen for loss-alignment and high-k sensitivity. Challenge point: if the conserved-quantity argument should be about KE in the inverse-cascade range, swap the weighting.

Calibration: weak sensor (recon/floor $1.8\times$ at Re$=1000$), monotone only on the noise ladder — kept as a cheap guard, weight 0.25.

5. Vorticity distribution — d_{W_1} (rewards.py:141)

The 1-Wasserstein distance between the sample's pointwise vorticity marginal F and the regime reference G, in the quantile representation

$$W_1(F,G) = \int_0^1 |F^{-1}(q) - G^{-1}(q)|\,\mathrm{d}q$$

discretized at $Q = 257$ evenly spaced ranks:

$$d_{W_1}(x) = \frac{1}{Q}\sum_{j=0}^{Q-1} |\ x_{\text{sorted}}[r_j] - q_{\text{ref}}(j)\ |, \qquad r_j = \operatorname{round}\left(j\frac{N^2-1}{Q-1}\right)$$

with x_{sorted} the ascending sort of x_{mid}'s N^2 pixels. (Sample side: nearest-rank order statistics; reference side: interpolated quantiles — a negligible asymmetry at $N^2 = 65536 \gg Q$.) Units: normalized vorticity.

Sensitive to PDF shape including the tails (the extreme-vorticity filaments that over-smoothing clips); blind to **where** values sit spatially, and to phases. Calibration: the weakest sensor — recons sit at $0.8\times$ the GT floor (the models do not measurably distort the marginal PDF on this task); kept as a guard (weight 0.25), candidate to drop. Challenge point: W_1 on velocity increments or on $|\nabla\omega|$ would target intermittency directly, which the pointwise marginal barely sees.

6. PDE residual — d_{pde}, $d_{\text{pde,lr}}$ (rewards.py:157; physics_guidance.py:38)

De-normalize $w = \sigma x + \mu$ and evaluate the Kolmogorov-flow vorticity equation residual at the middle frame (faithful port of the BaratiLab `voriticity_residual`, same operators as the data generator):

$$\mathcal{R}(w) = \underbrace{\partial_t\omega}_{\text{central FD}} + \underbrace{(u\cdot\nabla)\omega - \frac{1}{\mathrm{Re}}\Delta\omega}_{\text{spectral}} + \underbrace{0.1\omega}_{\text{drag}} + \underbrace{4\cos(4y)}_{\text{forcing}}$$

$$\partial_t\omega \approx \frac{w_3 - w_1}{2\Delta t}, \quad \Delta t = 1/32, \qquad u = (\partial_y\psi, -\partial_x\psi), \quad \Delta\psi = -\omega$$

Time derivative: central finite difference across the triplet (the only FD term). All spatial operators are spectral on $(0,2\pi)^2$ with integer wavenumbers; the velocity comes from the spectral Poisson solve for the streamfunction; **no** 2/3 dealiasing (matching the reference implementation). The last two terms are the `kf_2d` source terms moved to the left side. Assumption to challenge: the target regime shares this forcing ($k_f = 4$) and drag (0.1ω); Re is the only parameter varied.

$$d_{\text{pde}}(x) = \langle \mathcal{R}(w)^2 \rangle, \qquad d_{\text{pde,lr}}(x) = \left[\ln d_{\text{pde}}(x) - \ln r_{\text{ref}}\right]^2$$

The GT floor r_{ref} is not zero. Ground-truth data itself has $d_{\text{pde}} = 1.56/17.46/45.99$ at Re $= 500/1000/2000$ (measured), from $O(\Delta t^2)$ temporal truncation plus spectral truncation/aliasing of unresolved scales. Two calibration findings follow:

1. Raw d_{pde} mildly **rewards blurring**: mild smoothing filters the GT's own truncation noise (residual drops $25 \to 17$ at blur $\sigma_b \approx 1$) — a reward-hacking direction.
2. DDPO therefore uses $d_{\text{pde,lr}}$: the optimum is "as NS-consistent as the data itself", and leaving the floor in **either** direction (dirtier, or suspiciously cleaner than the physics allows) is penalized.

This is the only component that sees **dynamics/phases**: under spectrum-matched noise (all phases randomized, spectrum bit-exact) $d_{\text{pde,lr}}$ fires at $146\times$ its floor while all statistical components sit at $1.0\times$. Known wart: r_{ref} varies by sequence at Re$=2000$ ($\approx$ 32–46) — the anchor is noisy there; flagged to fix before Re$=2000$ DDPO.

7. Combined reward and DDPO usage (`make_ddpo_reward`, rewards.py:181)

$$r(x) = -\sum_i w_i \frac{d_i(x)}{s_i}$$

- **Scales s_i:** the standard deviation of d_i over the **step-0 reconstruction population** at that regime (what the frozen model produces before finetuning), from `reward_calibration.json`. Purely a units choice making the w_i comparable knobs.
- **Starting weights** (to revise after first runs): $w_{\text{spec}} = 0.5$, $w_{\text{spec,hi}k} = 1.0$, $w_{\text{energy}} = 0.25$, $w_{W_1} = 0.25$, $w_{\text{pde}} = 1.0$ (log-ratio mode).
- **Advantages must be per-input.** For K samples $x^{(1)}, ..., x^{(K)}$ of the same conditioning input:

$$A^{(j)} = \frac{r\left(x^{(j)}\right) - \bar{r}}{\text{std}_j\ r\left(x^{(j)}\right)}, \qquad \bar{r} = \frac{1}{K}\sum_j r\left(x^{(j)}\right)$$

This is not optional: all anchors are regime-level, so each input carries a systematic offset (its own GT's deviation from the regime mean). That offset is constant across samples of one input and cancels in A; pooled normalization leaves it as the dominant noise. Measured: pooled per-frame rank correlation with hi-k retention ≈ 0.1; within-input ≈ -0.7.

8. Summary of deliberate choices (attack surface)

choice	alternative not taken	rationale
enstrophy spectrum	KE spectrum ($\times\ 1/2k^2$)	weights the deficit band; loss-aligned
geometric-mean anchor $\Lambda(k)$	arithmetic mean	typical frame $\approx$ floor; kills the blur exploit
squared log-ratio per shell	linear ratio / L2 on spectra	equal weight per cascade decade
band $k \in [1, 96)$	full $[0, 128)$	$k = 0$ trivial; $k \geq 96$ GT-energy noise
pde as log-ratio to GT floor	raw residual $\to 0$	floor $\neq 0$; raw rewards smoothing
W_1 on pointwise ω marginal	increment / $\lvert\nabla\omega\rvert$ PDFs	simplicity; intermittency not covered
$\sigma = 4.7988$ for all regimes	per-regime normalization	model convention; anchors absorb it
middle frame for statistics	all 3 frames	matches `spectrum_fn` / metric conventions
per-input advantage baseline	pooled batch normalization	regime anchors $\Rightarrow$ per-input offsets

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Companion files: `docs/ddpo_reward_math.md` (this content as repo markdown) · `docs/energy_spectrum_metric.md` (spectral background) · `base_results/reward_calibration.ipynb` (evidence) · `base_results/reward_calibration.json` (scales, floors, weights).